HW7 - STAT 709

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All four exercises are required. Complete every part of each exercise.

Problem 1. Modes of convergence

Jun Shao, *Mathematical Statistics*, second edition, Chapter 1: Exercises 116 and 141 (required).

Exercise 1.116

Let $X, X_1, X_2, \ldots$ be random variables. Find an example for each of the following cases:

(a) $X_n \xrightarrow{p} X$, but $\{X_n\}$ does not converge to X a.s.

(b) $X_n \xrightarrow{p} X$, but $\{X_n\}$ does not converge to X in L_p for any $p > 0$.

(c) $X_n \xrightarrow{d} X$, but $\{X_n\}$ does not converge to X in probability (do not use Example 1.26).

(d) $X_n \xrightarrow{p} X$, but $\{g(X_n)\}$ does not converge to $g(X)$ in probability for some function g.

Sol.

(a)

(b)

(c)

(d)

Exercise 1.141

Suppose that X_n is a random variable having the binomial distribution $Bi(p_n, n)$. Show that if $np_n \to \theta > 0$, then $X_n \xrightarrow{d} X$, where X has the Poisson distribution $P(\theta)$.

Sol.

Problem 2. Asymptotic distributions and estimation

Bickel and Doksum, *Mathematical Statistics: Basic Ideas and Selected Topics*, Volume I, second edition, Chapter 5, Section 5.6, Problems for Section 5.3: Exercises 25 and 26 (required).

Exercise 5.3.25

Let $X_1, \ldots, X_n$ be a sample from a population with $\sigma^2 = \mathrm{Var}(X) < \infty$, $\mu = E(X)$ and let $T = \bar{X}^2$ be an estimate of μ^2.

(a) When $\mu \neq 0$, find the asymptotic distribution of $\sqrt{n}(T - \mu^2)$ using the delta method.

(b) When $\mu = 0$, find the asymptotic distribution of nT using

$$P(nT \leq t) = P\big(-\sqrt{t} \leq \sqrt{n}\bar{X} \leq \sqrt{t}\big).$$

Compare your answer to the answer in part (a).

(c) Find the limiting laws of $\sqrt{n}(\bar{X} - \mu)^2$ and $n(\bar{X} - \mu)^2$.

Sol.

(a)

(b)

(c)

Exercise 5.3.26

Show that if $X_1, \ldots, X_n$ are i.i.d. $\mathcal{N}(\mu, \sigma^2)$, then

$$\sqrt{n}\big(\bar{X} - \mu, \widehat{\sigma}^2 - \sigma^2\big) \xrightarrow{\mathcal{L}} \mathcal{N}(0, 0, \Sigma_0)$$

where $\Sigma_0 = \mathrm{diag}(\sigma^2, 2\sigma^4)$.

Hint: Use (5.3.33) and Theorem 5.3.4.

Sol.